EEL 6616 Take Home 3

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Problem 1

Part (a)

For the plant $P(z) = \frac{b}{z-a}$,

$$y(k+1) = bu(k) + f_1(k), \qquad f_1(k) = ay(k)$$

$$y(k+2) = bu(k+1) + abu(k) + f_2(k), \qquad f_2(k) = a^2y(k)$$

$$y(k+3) = bu(k+2) + abu(k+1) + a^2bu(k) + f_3(k), \qquad f_3(k) = a^3y(k)$$

$$\begin{bmatrix} y(k+1) \\ y(k+2) \\ y(k+3) \end{bmatrix} = G \begin{bmatrix} u(k) \\ u(k+1) \\ u(k+2) \end{bmatrix} + F$$

The matrices G and F are given by

$$G = \begin{bmatrix} b & 0 & 0 \\ ab & b & 0 \\ a^2b & ab & b \end{bmatrix} = \begin{bmatrix} 0.0952 & 0 & 0 \\ 0.0861 & 0.0952 & 0 \\ 0.0779 & 0.0861 & 0.0952 \end{bmatrix}, \qquad F = \begin{bmatrix} ay(k) \\ a^2y(k) \\ a^3y(k) \end{bmatrix} = \begin{bmatrix} 0.9048y(k) \\ 0.8187y(k) \\ 0.7407y(k) \end{bmatrix}$$

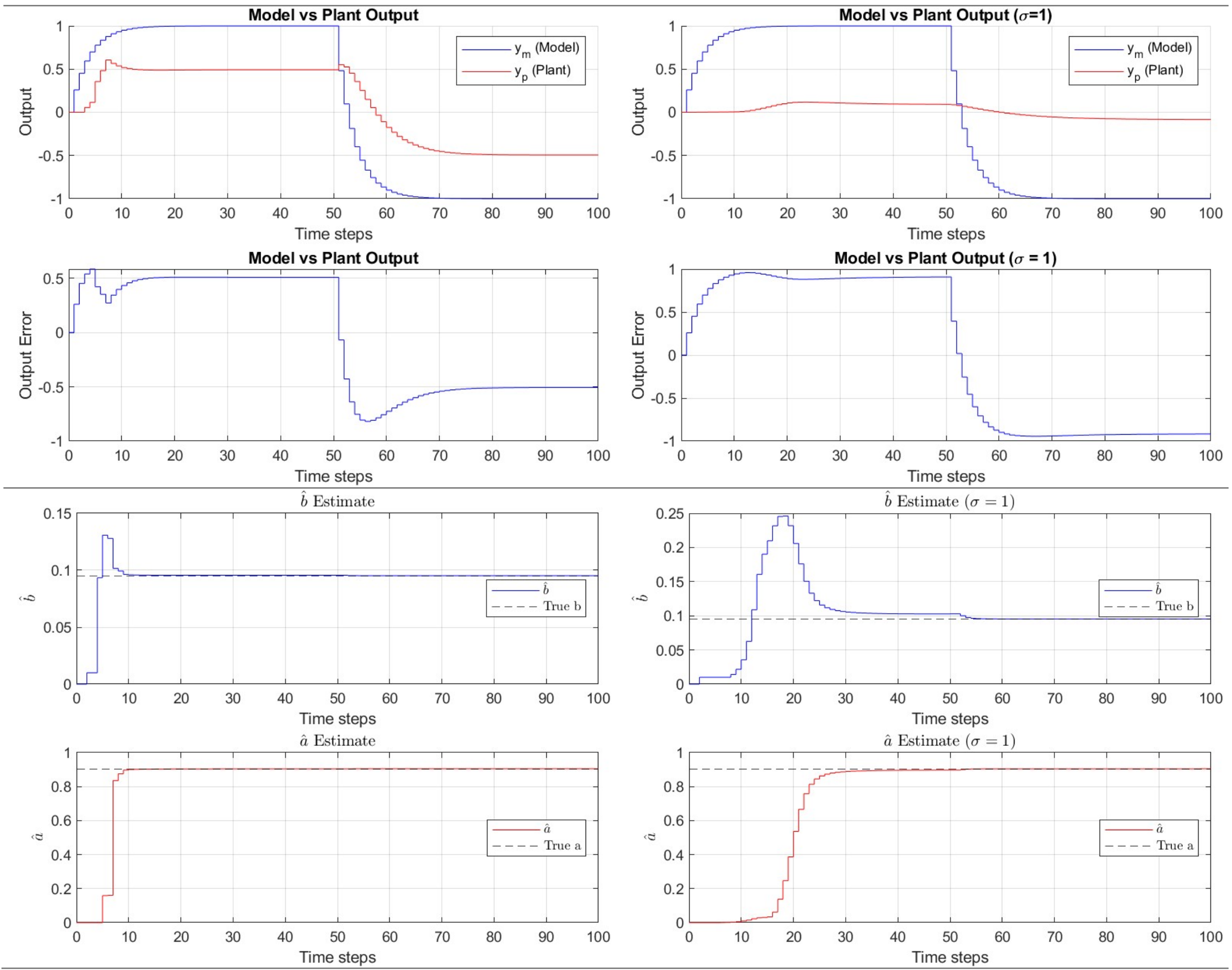
In both cases, $\hat{b} = 0$ means the expected plant gain is 0. The controller would expect the plant to be unresponsive and apply no input. The plant will be running on an open loop without zero input applied. In the adaptive controller, $\hat{b}(k) = 0$ drives the input $u(k) = 0$. With no input, the regressor functions of the RLSFF algorithm lose excitation and are unable to move $\hat{b}$ from 0, trapping the estimate. This can be avoided by freezing $\hat{b}$ before it reaches zero or preventing $\hat{b}$ from entering a neighborhood around 0. The problem statement specifies as strategy to modify $\hat{b}$:

$$\hat{b} \leftarrow \begin{cases} \hat{b}, & |\hat{b}| \geq \varepsilon \\ \varepsilon \operatorname{sign}(\hat{b}), & |\hat{b}| < \varepsilon \end{cases}$$

Part (b)

Figure 1 shows the results of the generalized predictive control with $\sigma = 0.01$ and $\sigma = 1$. It seems that the controller is able to control the plant such that the plant output behaves similar to the reference model output, although with some notable differences such as an apparent lag and steady-state errors. These both seem to increase in severity with σ. During optimization, σ represents the weight of the control input in the cost function. It serves as a tradeoff between control effort and tracking accuracy. The steady state error is a direct result of the presence of σ. Perfectly tracking the reference at steady state places a demand on control effort, which σ penalizes. The optimal solution thus has a constant offset at steady state, which grows with σ. The lag of y_p behind y_m results from the GPC failing to impose the model dynamics fast enough due to restrictions imposed by σ. The inputs strong enough to do so are discouraged by optimization.

$\boldsymbol{\sigma = 0.01}$	$\boldsymbol{\sigma = 1}$



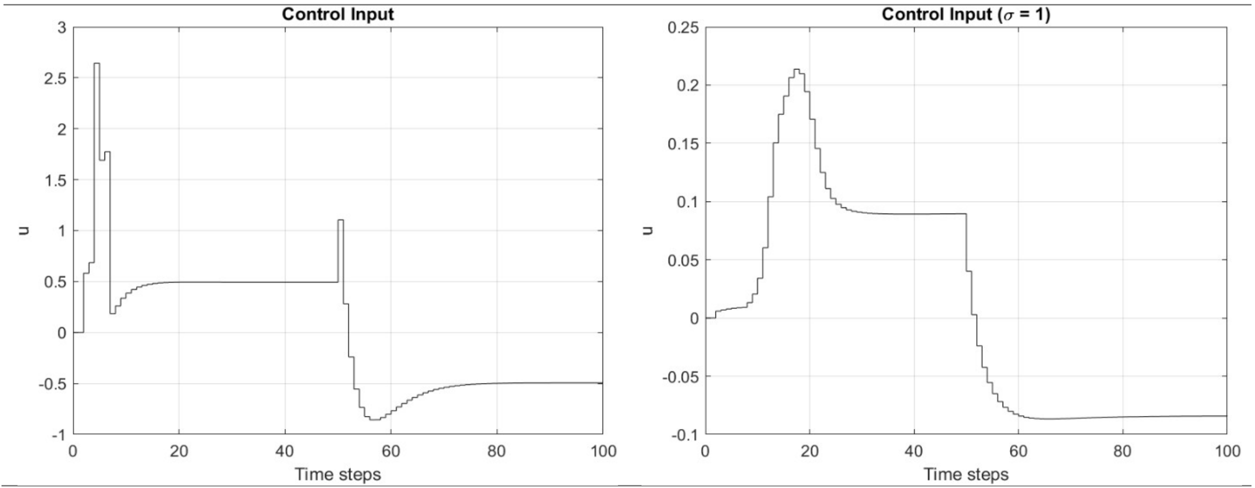


Figure 1: GPC Results

The parameter estimates suffered as a consequence of the reduced input strength. With weaker u values, the RLSFF regressor has less information, experiences less excitation, and convergence slows. With $\sigma = 1$, the estimator still reached the true values of a and b, but took 20-40 more time-steps than $\sigma = 0.01$. For this first order plant, persistent excitation of order 2 is sufficiently rich. The transient activity in u and y_p provide a rich enough signal, despite the flat reference input. The correct identification of parameters confirms this.

In both cases, $\hat{a}$ rose without overshoot while $\hat{b}$ overshot its nominal value, with the overshoot worsening for a larger σ. The initial growth in u and y_p during the transient response produce large updates in the parameters that cause $\hat{b}$ to jump past the true value.

Problem 2

$$M(s) = \frac{s+3}{(s+1)(s+5)}$$

Part (a)

$$P(s) = \frac{s+1}{s^2+1}$$

Both the plant and model are minimum phase and match in relative degree. This is a second order plant with relative degree of 1. This prescribes the degree of $\lambda(s)$ for a minimal order estimator to $\deg(\lambda) \geq 1$. $\lambda(s)$ has been selected as $\lambda(s) = s + 3$. In satisfying the conditions on the degree and monic requirements of control elements, the following are true: $q = 1$, $c^*(s)$ will also be a constant = c^*, and $d^*(s)$ will be of degree 1. The nominal value c_0^* is prescribed by the high-frequency gains of $M(s)$ and $P(s)$ such that $c_0^* = 1$. The solution for the nominal values c^*, d_0^*, and d^* is

$$\lambda(s) - c^*(s) = q(s)n_p(s) \rightarrow c^* = 2$$

$$-k_p d^*(s) = \lambda_0(s)d_m(s) - q(s)d_p(s), \qquad \frac{d^*(s)}{\lambda(s)} = d_0^* + \frac{d^*}{\lambda(s)}$$

$$\Rightarrow d^*(s) = -6s - 4, \qquad \frac{d^*(s)}{\lambda(s)} = -6 + \frac{14}{\lambda(s)}$$

The nominal values are then

$$c_0^* = 1, \qquad c^* = 2, \qquad d_0^* = -6, \qquad d^* = 14$$

If these controller parameters are held at their nominal values, the closed-loop transfer function of the system exactly matches the reference model $M(s)$.

The evolution of the output error and adaptation parameters are shown in the figures below. From Figure 2, the direct MRAC algorithm is able to control the plant such that the output error between the plant and model reference tends to zero. This is guaranteed from the direct MRAC and pseudo-gradient algorithm, as M(s) is SPR and $k_p = 1 > 0$.

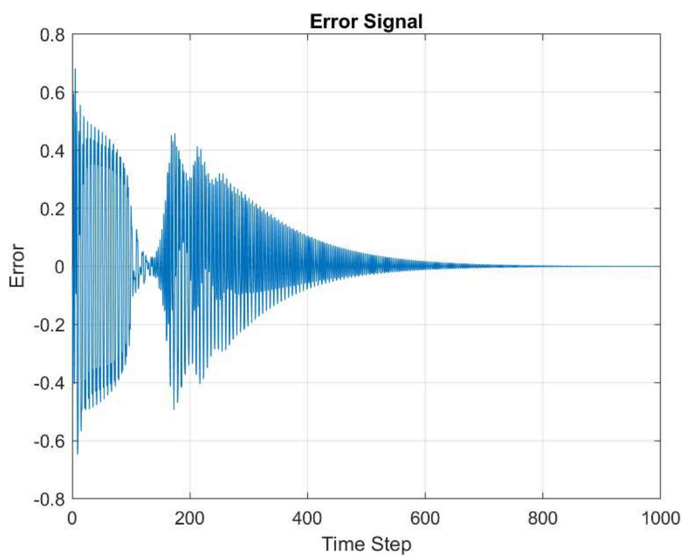


Figure 2: Part (a) Output Error

The parameters themselves are not approaching the nominal value. This seems to be a matter of excitation. The reference signal, $r(t)$, is a sine wave comprised of a single frequency. Thus, it provides persistent excitation of order 2. The direct MRAC implementation has 4 adaptive parameters, so unique parameter convergence requires persistent excitation of at least order 4. This would require an additional frequency to be present in $r(t)$. As is, the regressors do not have sufficient richness for the parameter convergence to be unique.

c_0 and d_0 show prominent oscillations because their corresponding regressor signals are the unfiltered, periodic signals r and y_p. The parameters c and d are smoother because their regressors are the filtered r and y_p signals.

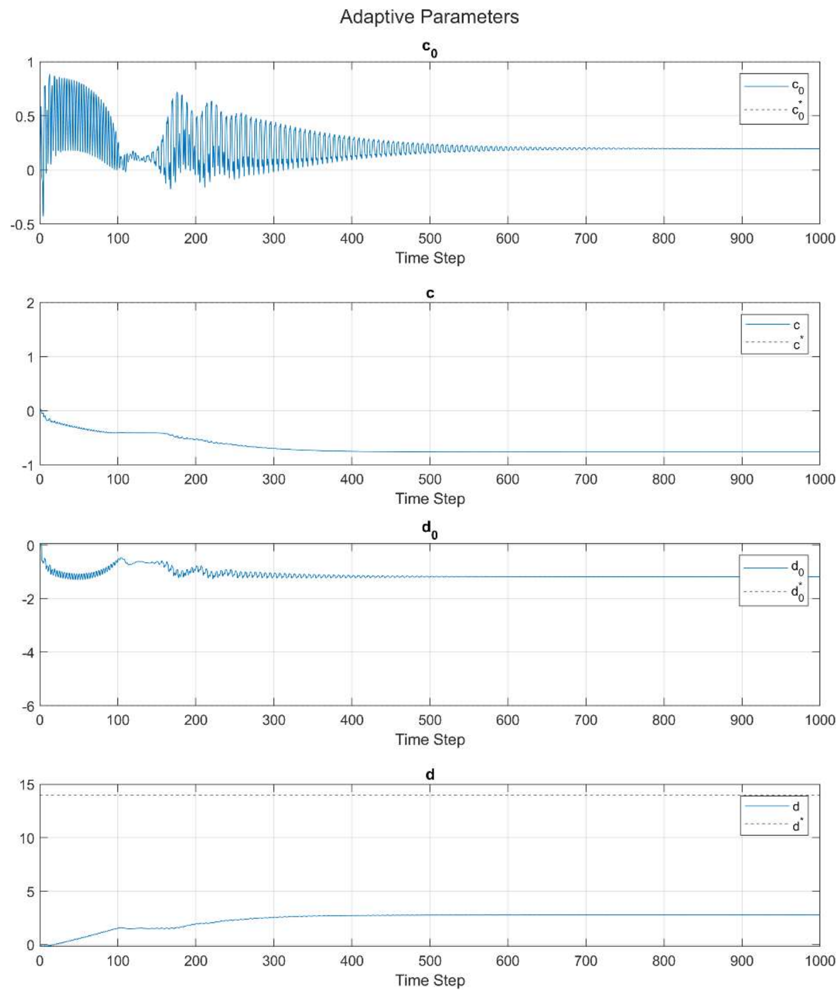


Figure 3: Part (a) Adaptive Parameter Evolution

Part (b)

$$P(s) = \frac{s+3}{(s-1)^2}$$

Although the plant has unstable poles, it is minimum phase with relative degree one, so direct MRAC can still be applied. The adaptive controller must simultaneously stabilize the plant and match the reference model. The closed-loop system may be unstable, raising some concerns about the boundedness of y_p as a regressor for the pseudo-gradient algorithm. These would need to be addressed to discuss error stability.

Using the solutions from Part (a), c_0^*, q, and $\lambda(s)$ remain unchanged.

$$\lambda(s) - c^*(s) = qn_p(s) \rightarrow c^* = 0$$

$$-k_p d^*(s) = d_m(s) - qd_p(s), \qquad \frac{d^*(s)}{\lambda(s)} = d_0^* + \frac{d^*}{\lambda(s)}$$

$$\Rightarrow d^*(s) = -8s - 4, \qquad \frac{d^*(s)}{\lambda(s)} = -8 + \frac{20}{\lambda(s)}$$

$$c^* = 0, \qquad d_0^* = -8, \qquad d^* = 20$$

The nominal controller parameters for this plant and model are

$$c_0^* = 1, \qquad c^* = 0, \qquad d_0^* = -8, \qquad d^* = 20$$

In this case, the plant and model have identical zeros, which is why $c^* = 0$.

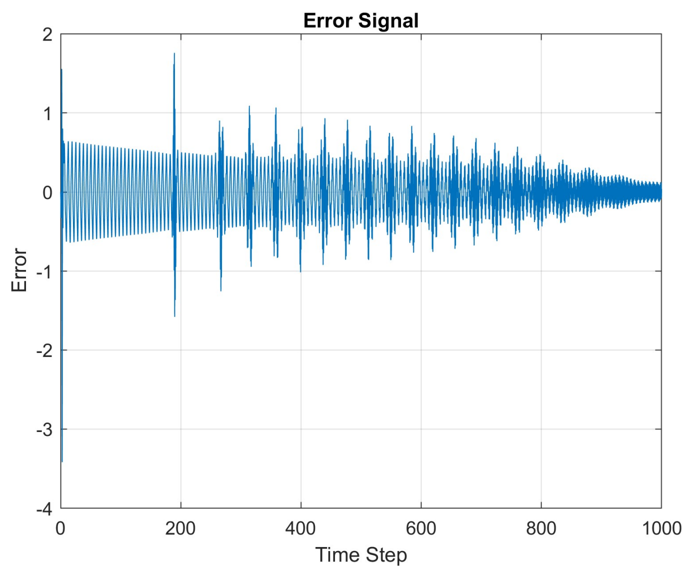


Figure 4: Part (b) Output Error

For this plant, the error does near 0, but convergence takes much longer than the stable plant. The observed error, shown in Figure 4, is also much higher. This is caused by instability in the transient, where parameters have not yet converged and the closed loop transfer function has not yet matched the reference model. Occasionally, there are large spikes in the error signal. These correspond to large jumps in the controller parameters, which can be seen in Figure 5.

Unlike part (a), the parameter d grows even through $t = 1000$. Again, this is caused by the lack of richness of the regressor signals. The other parameters show oscillations corresponding to the error spikes. c_0^* and d_0^* also exhibit smaller higher-frequency oscillations as well. The output error decays regardless.

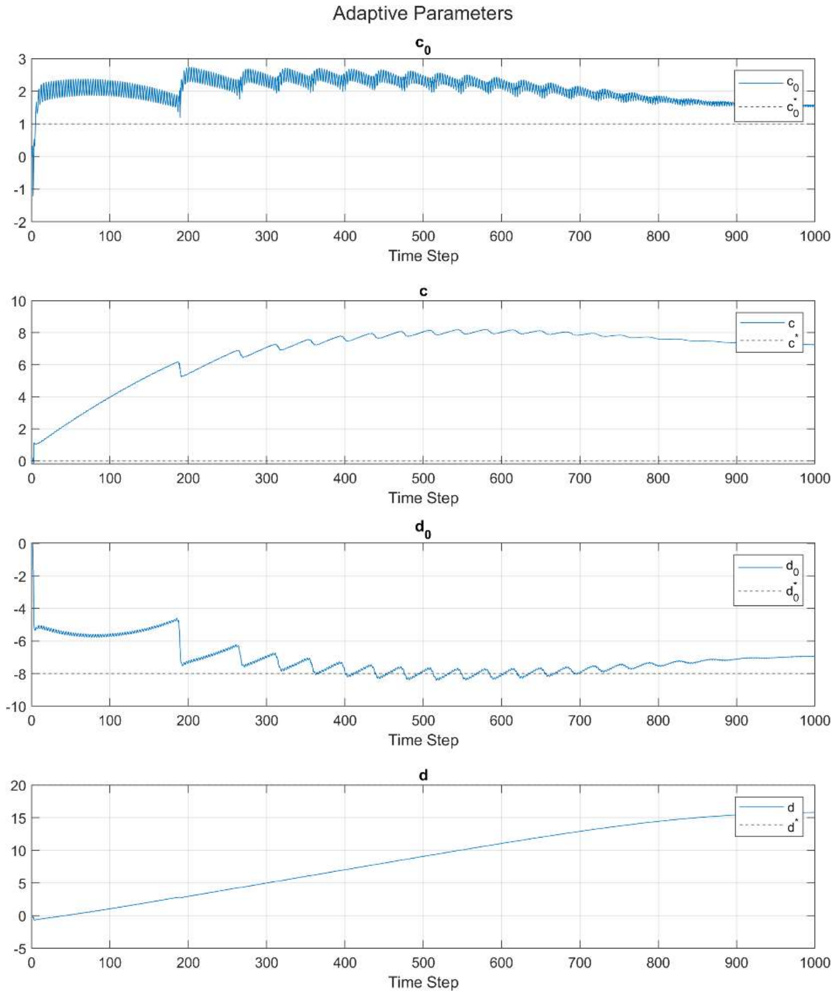


Figure 5: Part (b) Adaptive Parameter Evolution

Problem 3

$$P(z) = \frac{b}{z-a} = \frac{0.0952}{z-0.9048}, \qquad M(z) = \frac{b_m}{z-a_m} = \frac{0.2592}{z-0.7408}$$

Part (a)

The plant and model have the same relative degree and are both minimum phase. This means that indirect MRAC can be applied to this problem. For a minimal degree estimator and state variable filter, $\lambda(z) = z^{n-1}$. In the case of a first order system, $\lambda(z)$ is a constant term. Thus, the filter dynamics will not affect the controller algorithm. A constant state variable filter requires that $c(s)$ has degree -1 and $d(s)$ will be constant. $c(s)$ cannot be a polynomial, so $c(s) = 0$. The resulting transfer function from y_p to u is

$$U = c_0 R + \left(\frac{d}{\lambda} + d_0\right) Y_p$$

These constants can be rewritten with $d_0 \leftarrow \frac{d}{\lambda} + d_0$, yielding

$$u(k) = c_0 r(k) + d_0 y_p(k)$$

Higher degree $\lambda(z)$ with $\mathrm{degree}(\lambda) > 0$ would mean nontrivial c and d values, but lose minimal order. The nominal parameter for c_0 and d_0 are found from matching the reference and plant models

$$P_{cl}(z) = M(z)$$

$$\frac{bc_0}{z - a - bd_0} = \frac{b_m}{z - a_m}$$

$$\Rightarrow c_0 = \frac{b_m}{b} = 2.7227, \qquad d_0 = \frac{a_m - a}{b} = -1.7227$$

The expression presented does show singularities at $b = 0$, which causes an unbounded control input. This can be avoided by substituting b with $b + \varepsilon$ or defining a lower bound on b.

Indirect MRAC will be implemented by estimating plant parameters a and b. Defining these estimates as $\hat{a}$ and $\hat{b}$, the adaptive control algorithm is shown below. The parameter estimates are the results of the RLSFF algorithm.

$$u(k) = \frac{b_m}{\hat{b}} r(k) + \frac{a_m - \hat{a}}{\hat{b}} y_p(k)$$

$$c_0 = \frac{b_m}{\hat{b}}, \qquad d_0 = \frac{a_m - \hat{a}}{\hat{b}}$$

To account prevent singularities from causing issues,

$$\hat{b} \leftarrow \begin{cases} \hat{b}, & |\hat{b}| \geq \varepsilon \\ \varepsilon \,\mathrm{sign}(\hat{b}), & |\hat{b}| < \varepsilon \end{cases}$$

Part (b)

The plant output exhibits a large spike, reaching just above 35 during the transient response before quickly approaching the reference model output. After step 20, the plant and model output are almost the same. The reason for the spike during transience is the initial values of $\hat{b} = 0$. From the control law above, small values of $\hat{b}$ drive the control input to very large values. This in turn will cause the plant output to explode as it does during the early steps. This is mitigated by the freezing of $\hat{b}$ at small values.

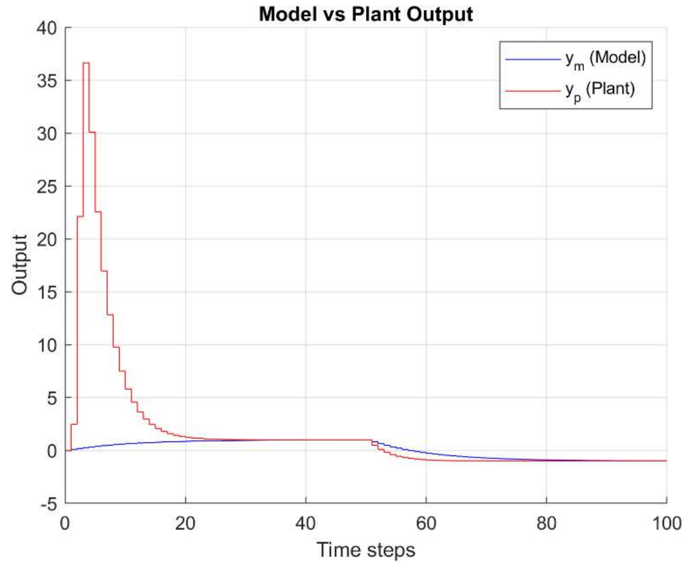


Figure 6: Indirect MRAC Output

After the initial transience, output error converges as the control law remains well defined, the closed-loop signals remain bounded, and the regressor signals persistently excite. When the reference switches from 1 to -1, the plant model does respond slightly faster than the reference model. This may indicate residual errors in the parameters.

As in Problem 1, the estimator is able to correctly identify plant parameters, meaning the regressors provided sufficiently rich persistent excitation. As the plant parameter estimates tend towards their true values, the controller parameters approach their nominal values at similar rates as seen in Figure and Figure . Thus, the closed-loop transfer function approach $M(z)$ and y_p tends to y_m.

Both $\hat{a}$ and $\hat{b}$ converge within 5 time steps, driven by the large transient spike in u and y_p, which produces large RLSFF updates. Since c_0 and d_0 are algebraic functions of $\hat{a}$ and $\hat{b}$, the controller parameters track the plant estimates at effectively the same rate.

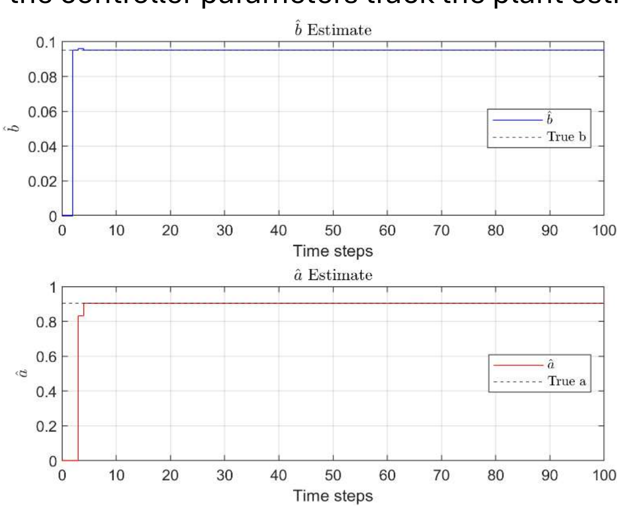


Figure 7: Plant Parameter Estimates

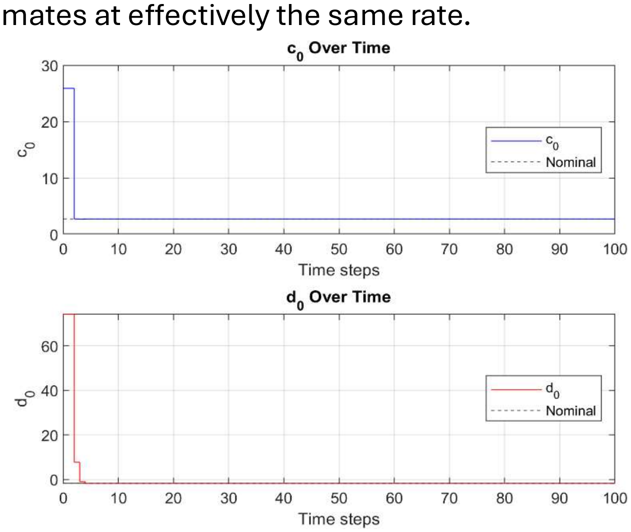


Figure 8: Controller Parameter Adaptation

Appendix: Simulink Models

The Simulink models and all images used in this report can be found at:
https://github.com/QuiteFiber93/eel6616/tree/main/takehome3